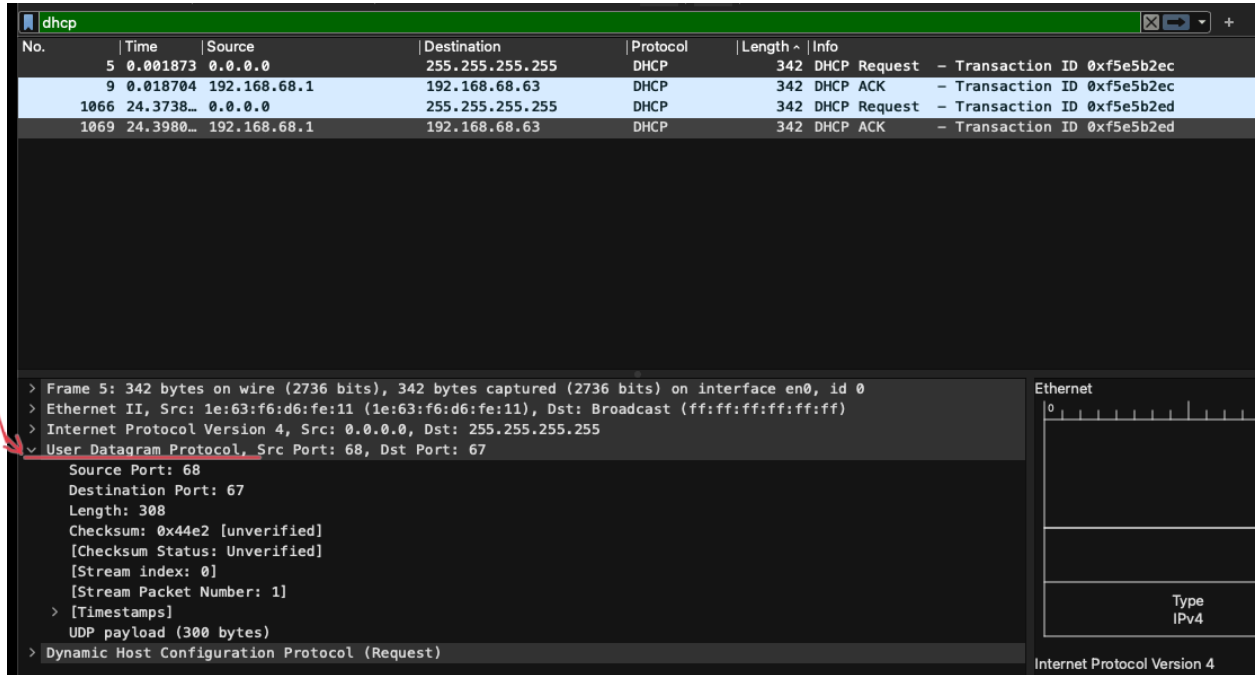


Lab 07 DHCP

Note: I tried to do it on my Mac OS, but for some questions, it didn't work, so I also used the provided trace file.

1. Are DHCP messages sent over UDP or TCP?

The DHCP messages sent over UDP



No.	Time	Source	Destination	Protocol	Length	Info
5	0.001873	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0xf5e5b2ec
9	0.018704	192.168.68.1	192.168.68.63	DHCP	342	DHCP ACK - Transaction ID 0xf5e5b2ec
1066	24.3738...	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0xf5e5b2ed
1069	24.3980...	192.168.68.1	192.168.68.63	DHCP	342	DHCP ACK - Transaction ID 0xf5e5b2ed

> Frame 5: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface en0, id 0
> Ethernet II, Src: 1e:63:f6:d6:fe:11 (1e:63:f6:d6:fe:11), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
User Datagram Protocol, Src Port: 68, Dst Port: 67
Source Port: 68
Destination Port: 67
Length: 308
Checksum: 0x44e2 [unverified]
[Checksum Status: Unverified]
[Stream index: 0]
[Stream Packet Number: 1]
> [Timestamps]
UDP payload (300 bytes)
> Dynamic Host Configuration Protocol (Request)

Ethernet
0
Type
IPv4
Internet Protocol Version 4

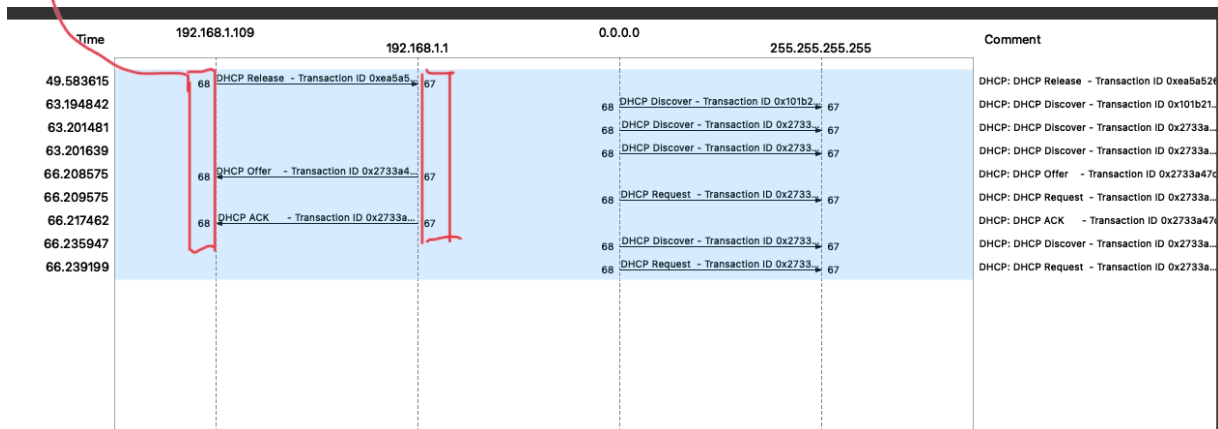
2. Draw a timing datagram illustrating the sequence of the first four-packet Discover/Offer/Request/ACK DHCP exchange between the client and server. For each packet, indicated the source and destination port numbers. Are the port numbers the same as in the example given in this lab assignment?

Src – 68, Dst – 67

Dst – 67, Src – 68

Src – 68, Dst – 67

Dst – 67, Src – 68

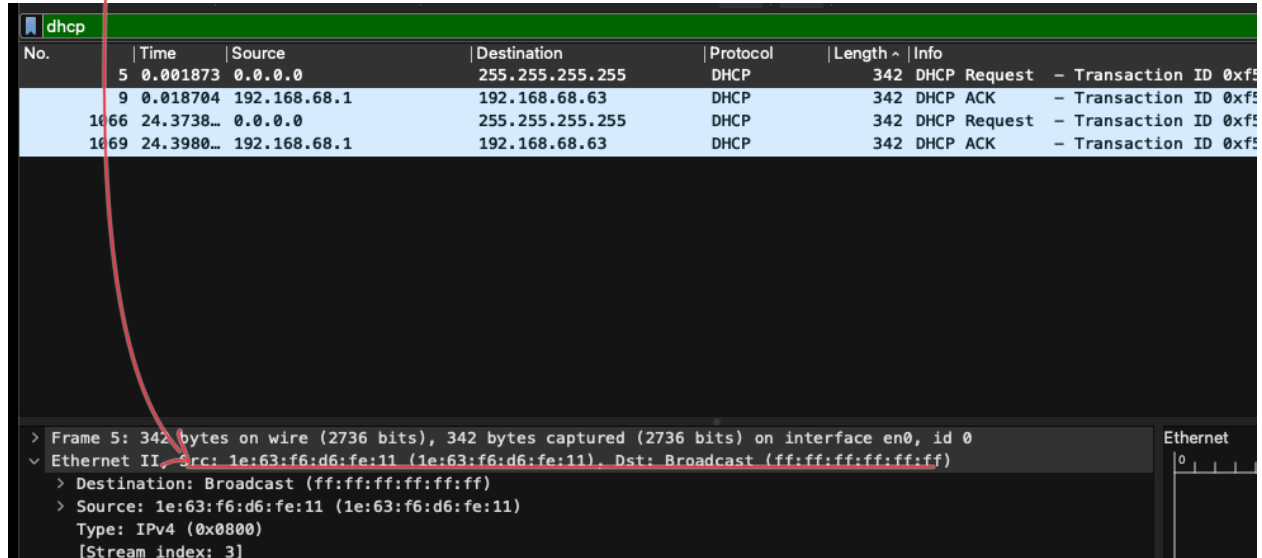


3. What is the link-layer (e.g., Ethernet) address of your host?

Ethernet II, Src: 1e:63:f6:d6:fe:11 (1e:63:f6:d6:fe:11)

Destination: Broadcast (ff:ff:ff:ff:ff:ff)

Source: 1e:63:f6:d6:fe:11 (1e:63:f6:d6:fe:11)



No.	Time	Source	Destination	Protocol	Length	Info
5	0.001873	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0xf5
9	0.018704	192.168.68.1	192.168.68.63	DHCP	342	DHCP ACK - Transaction ID 0xf5
1666	24.3738...	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0xf5
1669	24.3980...	192.168.68.1	192.168.68.63	DHCP	342	DHCP ACK - Transaction ID 0xf5

> Frame 5: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface en0, id 0

✓ Ethernet II, Src: 1e:63:f6:d6:fe:11 (1e:63:f6:d6:fe:11), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

> Destination: Broadcast (ff:ff:ff:ff:ff:ff)

> Source: 1e:63:f6:d6:fe:11 (1e:63:f6:d6:fe:11)

Type: IPv4 (0x0800)

[Stream index: 3]

4. What values in the DHCP discover message differentiate this message from the DHCP request message?

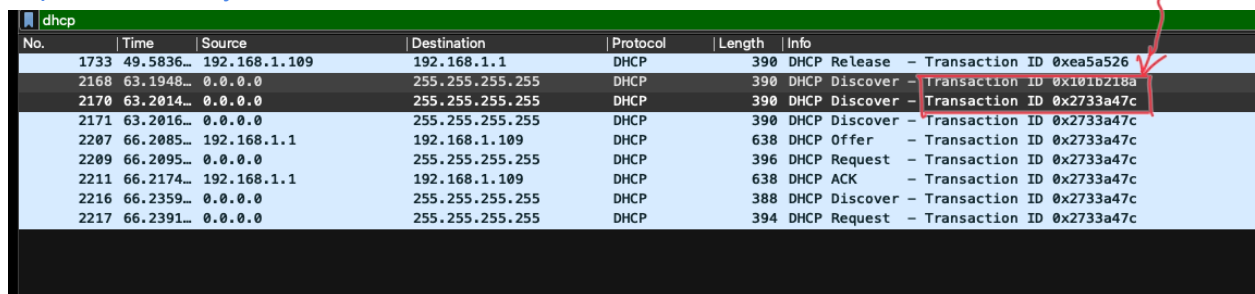
The message type value is different from the discover and request message. 1 is for discover and 3 is for request.

5. What is the value of the Transaction-ID in each of the first four (Discover/Offer/Request/ACK) DHCP messages? What are the values of the Transaction-ID in the second set (Request/ACK) set of DHCP messages? What is the purpose of the Transaction-ID field?

The transaction_ID of the first set of message: 0x101b218a

The transaction_ID of the 2nd set of message: 0x2733a47c

Purpose: The transaction ID is different so that the host can differentiate between different requests made by the user.



No.	Time	Source	Destination	Protocol	Length	Info
1733	49.5836...	192.168.1.109	192.168.1.1	DHCP	390	DHCP Release - Transaction ID 0xea5a526
2168	63.1948...	0.0.0.0	255.255.255.255	DHCP	390	DHCP Discover - Transaction ID 0x101b218a
2170	63.2014...	0.0.0.0	255.255.255.255	DHCP	390	DHCP Discover - Transaction ID 0x2733a47c
2171	63.2016...	0.0.0.0	255.255.255.255	DHCP	390	DHCP Discover - Transaction ID 0x2733a47c
2207	66.2085...	192.168.1.109	192.168.1.109	DHCP	638	DHCP Offer - Transaction ID 0x2733a47c
2209	66.2095...	0.0.0.0	255.255.255.255	DHCP	396	DHCP Request - Transaction ID 0x2733a47c
2211	66.2174...	192.168.1.1	192.168.1.109	DHCP	638	DHCP ACK - Transaction ID 0x2733a47c
2216	66.2359...	0.0.0.0	255.255.255.255	DHCP	388	DHCP Discover - Transaction ID 0x2733a47c
2217	66.2391...	0.0.0.0	255.255.255.255	DHCP	394	DHCP Request - Transaction ID 0x2733a47c

6. A host uses DHCP to obtain an IP address, among other things. But a host's IP address is not confirmed until the end of the four-message exchange! If the IP address is not set until the end of the four-message exchange, then what values are used in the IP datagrams in the four-message exchange? For each of the four DHCP messages (Discover/Offer/Request/ACK DHCP), indicate the source and destination IP addresses that are carried in the encapsulating IP datagram.

Discover: 0.0.0.0/255.255.255.255

Offer: 192.168.1.1/255.255.255.255 192.168.1.109

Request: 0.0.0.0/255.255.255.255

ACK: 192.168.1.1/255.255.255.255 192.168.1.109

2168	63.1948...	0.0.0.0	255.255.255.255	DHCP	390	DHCP Discover	- Transaction ID 0x101b218a
2170	63.2014...	0.0.0.0	255.255.255.255	DHCP	390	DHCP Discover	- Transaction ID 0x2733a47c
2171	63.2016...	0.0.0.0	255.255.255.255	DHCP	390	DHCP Discover	- Transaction ID 0x2733a47c
2207	66.2085...	192.168.1.1	192.168.1.109	DHCP	638	DHCP Offer	- Transaction ID 0x2733a47c
2209	66.2095...	0.0.0.0	255.255.255.255	DHCP	396	DHCP Request	- Transaction ID 0x2733a47c
2211	66.2174...	192.168.1.1	192.168.1.109	DHCP	638	DHCP ACK	- Transaction ID 0x2733a47c
2216	66.2359...	0.0.0.0	255.255.255.255	DHCP	388	DHCP Discover	- Transaction ID 0x2733a47c
2217	66.2391...	0.0.0.0	255.255.255.255	DHCP	394	DHCP Request	- Transaction ID 0x2733a47c

7. What is the IP address of your DHCP server?

The IP address of my DHCP server is 192.168.68.1

dhcp						
No.	Time	Source	Destination	Protocol	Length	Info
9	0.038699	192.168.68.1	192.168.68.63	DHCP	342	DHCP ACK - Transaction ID: 0xf5e5b391
4103	34.2533...	192.168.68.1	192.168.68.63	DHCP	342	DHCP ACK - Transaction ID: 0xf5e5b391
4	0.001245	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID: 0xf5e5b391
4098	34.1897...	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID: 0xf5e5b391


```

> Frame 9: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface en0, id 0
> Ethernet II, Src: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac), Dst: 1e:63:f6:d6:fe:11 (1e:63:f6:d6:fe:11)
> Internet Protocol Version 4, Src: 192.168.68.1, Dst: 192.168.68.63
> User Datagram Protocol, Src Port: 67, Dst Port: 68
> Dynamic Host Configuration Protocol (ACK)
  Message type: Boot Reply (2)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0xf5e5b391
  Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
  Client IP address: 0.0.0.0
  Your (client) IP address: 192.168.68.63
  Next server IP address: 0.0.0.0
  Relay agent IP address: 0.0.0.0

```

8. What IP address is the DHCP server offering to your host in the DHCP Offer message? Indicate which DHCP message contains the offered DHCP address.

The DHCP server offers my client 192.168.1.109, The offer message contains the DHCP address offered by the server.

2171	63.2016...	0.0.0.0	255.255.255.255	DHCP	390	DHCP Discover	- Transaction ID 0x2733a47c
2207	66.2085...	192.168.1.1	192.168.1.109	DHCP	638	DHCP Offer	- Transaction ID 0x2733a47c
2209	66.2095...	0.0.0.0	255.255.255.255	DHCP	396	DHCP Request	- Transaction ID 0x2733a47c
2211	66.2174...	192.168.1.1	192.168.1.109	DHCP	638	DHCP ACK	- Transaction ID 0x2733a47c
2216	66.2359...	0.0.0.0	255.255.255.255	DHCP	388	DHCP Discover	- Transaction ID 0x2733a47c
2217	66.2391...	0.0.0.0	255.255.255.255	DHCP	394	DHCP Request	- Transaction ID 0x2733a47c


```

> Bootp flags: 0x0000 (Unicast)
Client IP address: 0.0.0.0
Your (client) IP address: 192.168.1.109
Next server IP address: 0.0.0.0
Relay agent IP address: 0.0.0.0
Client MAC address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
Client hardware address padding: 00000000000000000000
Server host name not given
Boot file name not given
Magic cookie: DHCP
  Option: (53) DHCP Message Type (Offer)
    Length: 1
    DHCP: Offer (2)
  Option: (54) DHCP Server Identifier (192.168.1.1)
    Length: 4
    DHCP Server Identifier: 192.168.1.1
  Option: (51) IP Address Lease Time
    Length: 4
    IP Address Lease Time: 1 day (86400)

```

9. In the example screenshot in this assignment, there is no relay agent between the host and the DHCP server. What values in the trace indicate the absence of a relay agent? Is there a relay agent in your experiment? If so what is the IP address of the agent?

In the given example, the value 0.0.0.0 signifies the absence of a relay agent. Similarly, in my capture, the relay agent value is also 0.0.0.0, confirming that no relay agent was present in my case as well.

10. Explain the purpose of the router and subnet mask lines in the DHCP offer Message.

The subnet mask line specifies the subnet mask the client should use, defining the network's address range. The router line designates the default gateway, indicating where the client should send outgoing messages by default.

11. In the DHCP trace file noted in footnote 2, the DHCP server offers a specific IP

address to the client (see also question 8. above). In the client's response to the

What do we mean by "annotate"? If you hand in a paper copy, please highlight where in the printout you've found the answer and add some text (preferably with a colored pen) noting what you found in what you've highlight. If you hand in an electronic copy, it would be great if you could also highlight and annotate. first server OFFER message, does the client accept this IP address? Where in the client's RESPONSE is the client's requested address?

The client confirms acceptance of the offered IP address within the request message. After receiving an offer for 192.168.1.109, the client responds by explicitly requesting that same IP address in its DHCP Request message.

No.	Time	Source	Destination	Protocol	Length	Info
1733	49.5836...	192.168.1.109	192.168.1.1	DHCP	390	DHCP Release - Transaction ID 0xea5a526
2168	63.1948...	0.0.0.0	255.255.255.255	DHCP	390	DHCP Discover - Transaction ID 0x101b218a
2170	63.2014...	0.0.0.0	255.255.255.255	DHCP	390	DHCP Discover - Transaction ID 0x2733a47c
2171	63.2016...	0.0.0.0	255.255.255.255	DHCP	390	DHCP Discover - Transaction ID 0x2733a47c
2207	66.2085...	192.168.1.1	192.168.1.109	DHCP	638	DHCP Offer - Transaction ID 0x2733a47c
2209	66.2095...	0.0.0.0	255.255.255.255	DHCP	396	DHCP Request - Transaction ID 0x2733a47c
2211	66.2174...	192.168.1.1	192.168.1.109	DHCP	638	DHCP ACK - Transaction ID 0x2733a47c
2216	66.2359...	0.0.0.0	255.255.255.255	DHCP	388	DHCP Discover - Transaction ID 0x2733a47c
2217	66.2391...	0.0.0.0	255.255.255.255	DHCP	394	DHCP Request - Transaction ID 0x2733a47c


```

Next server IP address: 0.0.0.0
Relay agent IP address: 0.0.0.0
Client MAC address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
Client hardware address padding: 000000000000000000
Server host name not given
Boot file name not given
Magic cookie: DHCP
Option: (53) DHCP Message Type (Request)
  Length: 1
  DHCP: Request (3)
Option: (61) Client identifier
  Length: 7
  Hardware type: Ethernet (0x01)
  Client MAC address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
Option: (50) Requested IP Address (192.168.1.109)
  Length: 4
  Requested IP Address: 192.168.1.109
Option: (54) DHCP Server Identifier (192.168.1.1)
  
```

12. Explain the purpose of the lease time. How long is the lease time in your experiment?

The lease time defines the duration for which a client can use the assigned IP address before requiring renewal or reassignment by the server. In my experiment, the lease time is 86,400 seconds (1 day).

The image shows a Wireshark packet capture of a DHCP transaction. The packet list at the top shows several packets, including a DHCP Offer (638) and a DHCP Request (396). The packet details pane on the right shows the structure of the DHCP Offer packet, including the Header revision (0), Header padding (0), Flags (0x10), Data rate (1 Mb), Channel flags (0x00a0), and Signal Quality (100). The packet bytes pane on the left shows the raw data of the packet, with the IP Address Lease Time field highlighted in red.

```

Seconds elapsed: 0
> Bootp flags: 0x0000 (Unicast)
Client IP address: 0.0.0.0
Your (client) IP address: 192.168.1.109
Next server IP address: 0.0.0.0
Relay agent IP address: 0.0.0.0
Client MAC address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
Client hardware address padding: 00000000000000000000
Server host name not given
Boot file name not given
Magic cookie: DHCP
Option: (53) DHCP Message Type (ACK)
  Length: 1
  DHCP: ACK (5)
Option: (54) DHCP Server Identifier (192.168.1.1)
  Length: 4
  DHCP Server Identifier: 192.168.1.1
Option: (51) IP Address Lease Time
  Length: 4
  IP Address Lease Time: 1 day (86400)
Option: (1) Subnet Mask (255.255.255.0)
  Length: 4
  Subnet Mask: 255.255.255.0
  
```

13. What is the purpose of the DHCP release message? Does the DHCP server issue an acknowledgment of receipt of the client's DHCP request? What would happen if the client's DHCP release message is lost?

The DHCP release message allows the client to return its assigned IP address to the server. However, the server does not provide any acknowledgment upon receiving this message. If the release message is lost, the client still relinquishes the IP address, but the server will not reassign it until the lease period expires.

14. Clear the bootp filter from your Wireshark window. Were any ARP packets sent or received during the DHCP packet-exchange period? If so, explain the purpose of those ARP packets.

Yes, ARP packets were observed during the DHCP packet exchange. These broadcasts are sent by the network to map known IP addresses to corresponding MAC addresses, helping clients establish communication within the network.

2202	65.8233_	CiscoLinksys_f7:1d:51	Broadcast	802.11	183	Beacon frame, SN=3755, FN=0, Flags=.....C, BI=100, SSID="30
2203	65.9257_	CiscoLinksys_f7:1d:51	Broadcast	802.11	183	Beacon frame, SN=3756, FN=0, Flags=.....C, BI=100, SSID="30
2204	66.0281_	CiscoLinksys_f7:1d:51	Broadcast	802.11	183	Beacon frame, SN=3757, FN=0, Flags=.....C, BI=100, SSID="30
2205	66.1305_	CiscoLinksys_f7:1d:51	Broadcast	802.11	183	Beacon frame, SN=3758, FN=0, Flags=.....C, BI=100, SSID="30
2206	66.2000_		Intel_d1:b6:4f	802.11	38	Acknowledgement, Flags=.....C
2207	66.2085_	192.168.1.1	192.168.1.109	DHCP	638	DHCP Offer - Transaction ID 0x2733a47c
2208	66.2089_		CiscoLinksys_f7:1d:51	802.11	38	Acknowledgement, Flags=.....C
2209	66.2095_	0.0.0.0	255.255.255.255	DHCP	396	DHCP Request - Transaction ID 0x2733a47c
2210	66.2096_		Intel_d1:b6:4f	802.11	38	Acknowledgement, Flags=.....C
2211	66.2174_	192.168.1.1	192.168.1.109	DHCP	638	DHCP ACK - Transaction ID 0x2733a47c
2212	66.2177_		CiscoLinksys_f7:1d:51	802.11	38	Acknowledgement, Flags=.....C
2213	66.2247_	Intel_d1:b6:4f	Broadcast	ARP	98	ARP Announcement for 192.168.1.109
2214	66.2248_		Intel_d1:b6:4f	802.11	38	Acknowledgement, Flags=.....C
2215	66.2328_	CiscoLinksys_f7:1d:51	Broadcast	802.11	183	Beacon frame, SN=3761, FN=0, Flags=.....C, BI=100, SSID="30
2216	66.2359_	0.0.0.0	255.255.255.255	DHCP	388	DHCP Discover - Transaction ID 0x2733a47c
2217	66.2391_	0.0.0.0	255.255.255.255	DHCP	394	DHCP Request - Transaction ID 0x2733a47c
2218	66.2400_	Intel_d1:b6:4f	Broadcast	ARP	88	ARP Announcement for 192.168.1.109
2219	66.3353_	CiscoLinksys_f7:1d:51	Broadcast	802.11	183	Beacon frame, SN=3762, FN=0, Flags=.....C, BI=100, SSID="30
2220	66.4376_	CiscoLinksys_f7:1d:51	Broadcast	802.11	183	Beacon frame, SN=3763, FN=0, Flags=.....C, BI=100, SSID="30
2221	66.5122_	Intel_d1:b6:4f	Broadcast	ARP	98	ARP Announcement for 192.168.1.109
2222	66.5123_	Intel_d1:b6:4f	Broadcast	ARP	98	ARP Announcement for 192.168.1.109
2223	66.5124_		Intel_d1:b6:4f	802.11	38	Acknowledgement, Flags=.....C
2224	66.5400_	CiscoLinksys_f7:1d:51	Broadcast	802.11	183	Beacon frame, SN=3764, FN=0, Flags=.....C, BI=100, SSID="30
2225	66.5403_	Intel_d1:b6:4f	Broadcast	ARP	88	ARP Announcement for 192.168.1.109
Dynamic Host Configuration Protocol (ACK)						
Message type: Boot Reply (2)						
Hardware type: Ethernet (0x01)						
Hardware address length: 6						

IEEE 802.11
0
Header